SatQuery AI — Where Things Stand

A plain-language status check, written 2026-09-05, ~2 days before the internal hackathon round.

1. What's fully done

Think of the project as 9 building steps. Steps 1 through 6 are completely finished and tested:

- **Steps 1-4** — reading satellite image files correctly, catching broken/mismatched images, fixing radar images so they're actually viewable (not just black), and a system that picks the right AI tool for a given question.
- **Step 5** — the "brain": a 9-step pipeline that takes a question + image(s), decides what kind of question it is, picks a tool, runs it, and explains its own reasoning at every step.
- **Step 6** — the actual website: drag in images, type a question, and watch the reasoning steps appear live on screen before the final answer shows up.

2. Step 7 — partially done

Step 7 is about training a real AI model instead of a placeholder. This is done, but on a dataset that turned out not to be the exact one the hackathon names (more on that in section 3).

- Trained a model that looks at a satellite patch and predicts what kind of land it shows (forest, farmland, water, urban area, etc. — 19 categories total).
- Trained it 3 ways: using only optical (camera-like) images, using only radar images, and using both together ("fusion"). This matters because one of the hackathon's required capabilities is proving that combining optical + radar is better than using just one.
- Result: the combined (fusion) version scored better than either one alone — which is exactly the proof that capability requires.
- This is wired into the real website already and verified working live — not just in theory.

The numbers (for reference):

Version	Score (higher = better)	Score (2nd measure)
Optical only	0.704	0.350
Radar only	0.658	0.285
Optical + Radar combined	0.705	0.368

3. What we just found out — the official dataset

You pasted the hackathon's actual brief. It names one specific dataset: **"BigEarthNet.txt"**. We checked it directly (the paper and its download page). Here's what it actually is, in plain terms:

- **It is NOT satellite images.** It's about 9.6 million pieces of written text.
- That text is: image captions ("this patch shows a river next to farmland"), question-and-answer pairs ("is there water here? yes"), and "point-to-this-part-of-the-image" instructions.

- It links to satellite images by an ID number, but the images themselves are a separate, much bigger download (the full BigEarthNet archive, ~110 GB) — your own project notes already flagged that as too big to fit in a free Kaggle/Colab session.

In short: the officially-named dataset is for teaching an AI to **describe and answer questions about** an image in writing. What we trained in Step 7 instead teaches an AI to **classify** an image into a category (forest/farmland/etc.) — a different, real, and useful skill, but not the exact one the brief names.

The good news: both datasets come from the same BigEarthNet family and use the same image ID system, so the already-downloaded images can likely be reused — we would just add the official text file on top, not start over from zero.

4. What's left to do

- **Use the official "BigEarthNet.txt" dataset** to replace the fake placeholder answers for "describe this image" / "what is in this image" type questions with real, trained ones. Currently those questions still get a canned sentence, not a real AI answer.
- **Step 8** — a real change-detection model (currently uses simple pixel-difference math instead of a trained AI, for before/after image comparisons).
- **Step 9** — making the live demo unbreakable (saved backup run in case wifi/internet fails on stage) and preparing presentation slides/assets.
- **Optional but recommended by your own notes:** register on Bhoonidhi (ISRO's own real data website) and test with a couple of genuine ISRO satellite images, since judges may test with real ISRO data, not just the training benchmark images.

5. The decision that's actually in front of you

Given there are roughly 2 days left, here are the realistic choices — pick one and we go:

- **Option A — Add the official dataset on top (recommended).** Keep the land-cover model that's already working and proven, and additionally wire in "BigEarthNet.txt" to fix the fake-answer problem for description/question-answering. More work, but nothing done so far gets thrown away, and it covers both required capabilities.
- **Option B — Start over, official dataset only.** Most exactly matches the brief's wording, but throws away the model you already trained and verified, and is risky this close to the deadline.
- **Option C — Keep what's built, be upfront about it.** Fastest and safest for the demo. State plainly in the presentation that a real BigEarthNet image subset was used for the fusion proof, and that the official text dataset is planned as a next step. The question-answering feature stays a placeholder for now.

Tell me which option (A, B, or C) — or ask me anything about any section above before deciding.